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DIALOGUES AS A FOUNDATION FOR INTUITIONISTIC LOGIC

SUMMARY OF CONTENTS

The principal content of this article is a (new) foundation for intuitionistic logic, based on an analysis of argumentative processes as codified in the concepts of a *dialogue* and a *strategy* for dialogues. This work is presented in Section 3. A general historical introduction is given in Section 2. Since already there the reader will need to know exactly what a dialogue and a strategy shall be, these basic concepts are defined in the (purely technical) Section 1.

1 BASIC CONCEPTS: DIALOGUES AND STRATEGIES

I consider a first-order language, built with variables $x, y, \dots$ and terms t ; formulas shall be constructed from atomic formulas with the propositional connectives $\wedge, \vee, \rightarrow, \neg$ and the quantifiers $\forall, \exists$; I shall also consider $\vee, \wedge_1, \wedge_2, \exists$ as *special symbols* in their own right. By an *expression* I understand either a term or a formula or a special symbol. I introduce two further symbols P and Q ; taking two new (and disjoint) copies of the set of expressions, I form for every expression e two new expressions Pe and Qe , the *P-signed* and the *Q-signed version* of the expression e .

The symbols P, Q shall symbolise two persons engaged in an argument or in a dialogue; I shall use X, Y as variables for P, Q and shall assume $X \neq Y$. An *argumentation form* is a schematic presentation of an argument, concerning a logically composite assertion; it describes how a composite assertion made by C may be *attacked* by Y and how, if possible, this attack may be *answered* by X . As the logical form of the composite assertion shall completely determine the argument, each of the four propositional connectives and each of the two quantifiers determines an argumentation form:

$\wedge$:	assertion:	$Xw_1 \wedge w_2$	
	attack:	$Y\wedge_i$	(i.e., Y chooses $i = 1$ or $i = 2$)
	answer:	Xw_i	
$\vee$:	assertion:	$Xw_1 \vee w_2$	
	attack:	$Y\vee$	
	answer:	Xw_i	(i.e., X chooses $i = 1$ or $i = 2$)
$\rightarrow$:	assertion:	$Xw_1 \rightarrow w_2$	
	attack:	Yw_1	
	answer:	Xw_2	
$\neg$:	assertion:	$x\neg w$	
	attack:	Yw	
	answer:	<i>no answer possible</i>	
$\forall$:	assertion:	$X\forall xw$	
	attack:	Yt	(i.e., Y chooses the term t)
	answer:	$Xw(t)$	
$\exists$:	assertion:	$X\exists xw$	
	attack:	$Y\exists$	
	answer:	$Xw(t)$	(i.e., X chooses the term t).

In the last two answers I have written $w(t)$ for the substitution instance obtained from w if the term t is substituted for the variable x .

A dialogue shall be a (finite or infinite) sequence δ of statements, i.e., signed expressions, stated alternately by P and Q and progressing in accordance with the argumentation forms; I shall consider only such dialogues which are begun by P . Since it is necessary to distinguish carefully between attacks, answers and the assertions they refer to, I shall introduce besides δ an accompanying sequence η of references, and there I shall use the symbols A for *attack* and D for *answer (defense)*. For notational convenience, I shall assume that a natural number *is* the set of all smaller natural numbers (whence 0 is the first natural number), and a *sequence* shall always be a function, defined on either a natural number or on the set ω of all natural numbers. The precise definition then reads as follows:

A *dialogue* δ, η consists of two sequences such that

δ is a sequence of signed expressions,

η is a function defined on the *positive* members of $\text{def}(\delta)$, and if n in $\text{def}(\eta)$ is an ordered pair $[m, Z]$ such that m is a natural number less than n and Z is either A or D ,

satisfying the properties (D00)–(D02):

- (D00) $\delta(n)$ is P -signed if n is even and Q -signed if n is odd; $\delta(0)$ is a composite formula.
- (D01) If $\eta(n) = [m, A]$ then $\delta(m)$ is a composite formula and $\delta(n)$ is attack upon $\delta(m)$ according to the appropriate argumentation form.
- (D02) If $\eta(p) = [n, D]$ then $\eta(n) = [m, A]$ and $\delta(p)$ is the answer to the attack $\delta(n)$ according to the appropriate argumentation form.

The signed formulas occurring as values of δ are called the *assertions* of the dialogue while the remaining values of δ are *symbolic statements* or, more correctly, *symbolic attacks*. The numbers in $\text{def}(\delta)$ are called the *positions* or *places* of the dialogue. If Pv is the assertion $\delta(0)$, the dialogue is said to be a dialogue *for* the formula v (or, sometimes, for Pv).

Assume now that a particular class H of dialogues is given, defined maybe by additional conditions, which has the property that, for every position n of an H -dialogue δ, η , the *restrictions* of δ, η to positions i such that $i \leq n$ form an H -dialogue again. Assume further that a subclass of H has been defined, consisting of certain *finite* H -dialogues which then are said to be the H -dialogues *won* by P . Let v be a composite formula; to say that P has an H -strategy shall mean that P is in possession of a system of information, consisting of possible choices of P -statements in dialogues, such that every H -dialogue for v is won by P if only P chooses, after every statement made by Q , its own statement from this system of information. In order to formulate a more precise definition, recall that a *tree* S is a partially ordered set of elements called *nodes* with the following properties: there exists a *largest* element e_S (the top node), and for every node e the number $\|e\|$ of nodes f such that $e \leq f < e_S$ is *finite*; every node except e_S has exactly one *upper neighbour* but may have arbitrarily many *lower neighbours* (i.e., the tree is branching downwards). A *path* in S is a linearly ordered subset of nodes which, together with each of its elements e , contains all the preceding nodes f with $e \leq f$; a *branch* is a path which is maximal. If A is a branch of S , let α_A be the unique order-preserving bijection which maps either a natural number or all of ω onto A , i.e. $\|\alpha_A(i)\| = i$ holds for every node $\alpha_A(i)$ in A . Consider now a tree S and functions δ, η where δ is defined on all nodes of S and η on the nodes different from e_S ; for every branch A define $\delta_A = \delta \cdot \alpha_A, \eta_A = \eta \cdot \alpha_A$. The triplet S, δ, η then is an H -strategy for v if

- (S0) For every branch A of S the pair δ_A, η_A is an H -dialogue for v which is won by P .
- (S1) For every node e of S the following is the case. If $\|e\|$ is odd then S does not branch at e . If $\|e\|$ is even then e has as many lower neighbours as Q has possibilities to extend, by adding a new position, to an H -dialogue the (restricted) dialogue leading to e ,

and δ, η assign these lower neighbours the values which realise these possibilities.

The general definitions having been established, particular classes of dialogues can be introduced. To do so, I shall need the following terminology. Let δ, η be a dialogue, and let $\delta(n)$ be one of its attacks. The attack $\delta(n)$ will be said to be *open at a position k* with $n < k$ if there is no position n' with $n < n' \leq k$ which carries an answer $\delta(n')$ to that attack. In particular, an attack upon a formula $X \neg v$ remains open at all later places. A *D-dialogue* shall be a dialogue δ, η satisfying the following properties (D10)–(D13) :

- (D10) P may assert an atomic formula only after it has been asserted by Q before: if $\delta(n) = Pa$ and a is atomic then there exists m such that $m < n$ and $\delta(m) = Qa$.
- (D11) If, at a position $p-1$, there are several open attacks suitable to be answered at p , then only the *latest* of them may be answered at p : if $\eta(p) = [n, D]$ and if $n < n' < p, n' - n \equiv 0 \pmod{2}$, $\eta(n') = [m', A]$ then there exists p' such that $n' < p' < p, \eta(p') = [n', D]$.
- (D12) An attack may be answered at most once: for every n there exists at most one p such that $\eta(p) = [n, D]$.
- (D13) A P -formula may be attacked at most once: if m is even then there exists at most one n such that $\eta(n) = [m, A]$.

A *D-dialogue* is said to be *won by P* if it is finite, ends with an even position and if the rules do not permit Q to continue with another attack or answer. In that case the last position carries an atomic formula asserted by P .

The importance of *D-dialogues* rests in the fact that the formulas for which there exist *D-strategies* are precisely those provable in intuitionistic logic. This follows from the following, stronger

EQUIVALENCE THEOREM. *There exist recursive algorithms which, for every formula v , transform a proof of the sequent $\Rightarrow v$ in Gentzen's calculus LJ (for intuitionistic logic) into a D-strategy — and vice versa.*

Contrary to first appearances, a proof of this theorem is by no mean obvious; it cannot be pursued here and may be found in Felscher [1981; 1985].

An *E-dialogue* shall be a *D-dialogue* satisfying the additional condition that Q can react only upon the immediately preceding utterance of P :

- (E) For every n in $\text{def}(\delta)$: if n is odd then $\delta(n)$ is either attack upon $\delta(n-1)$ or answer to $\delta(n-1)$.

An *E-dialogue* is said to be *won by P* if, again, it is finite, ends with an even position and if now the rules for *E-dialogues* do not permit Q to continue

with either an attack or an answer. There will be occasion to refer to the following result which is auxiliary to the proof of the Equivalence Theorem.

EXTENSION LEMMA. *There is a recursive algorithm by which every E -strategy can be embedded into a D -strategy.*

It follows from this lemma that the Equivalence theorem holds also for E -strategies in place of D -strategies.

Readers not familiar with the use of dialogues may appreciate the following *examples* in which $a, b, \dots$ are assumed to be atomic formulas.

(1a)

0.	$P(a \wedge b) \rightarrow (a \wedge b)$	
1.	$Q(a \wedge b)$	$[0, A]$
2.	$P \wedge_1$	$[1, A]$
3.	Qa	$[2, D]$
4.	$P \wedge_2$	$[1, A]$
5.	Qb	$[4, D]$
6.	$P(a \wedge b)$	$[1, D]$
7.	$Q \wedge_1$	$[6, Q]$
8.	Pa	$[7, D]$
7.	$Q \wedge_2$	$[6, Q]$
8.	Pb	$[7, D]$

(1b)

0.	$P(a \wedge b) \rightarrow (a \wedge b)$	
1.	$Q(a \wedge b)$	$[0, A]$
2.	$P(a \wedge b)$	$[1, D]$
3.	$Q \wedge_1$	$[2, A]$
4.	$P \wedge_1$	$[1, A]$
5.	Qa	$[4, D]$
6.	Pa	$[3, D]$
3.	$Q \wedge_2$	$[2, A]$
4.	$P \wedge_2$	$[1, A]$
5.	Qb	$[4, D]$
6.	Pb	$[3, D]$

Here we have two different D -strategies for the same formula.

(2a)

0.	$P(a \rightarrow \neg \neg a)$	
1.	Qa	$[0, A]$
2.	$P \neg \neg a$	$[1, D]$
3.	$Q \neg a$	$[2, A]$
4.	Pa	$[3, A]$

(2b)

0.	$P(\neg \neg a \rightarrow a)$	
1.	$Q \neg \neg a$	$[0, A]$
2.	$P \neg a$	$[1, A]$
3.	Qa	$[3, A]$

The first example is a D -strategy. In the second example, P cannot win if $(D11)$ shall not be violated.

(3)

- | | | |
|----|--------------------------------------|----------|
| 0. | $P((a \wedge \neg a) \rightarrow b)$ | |
| 1. | $Q(a \wedge \neg a)$ | $[0, A]$ |
| 2. | $P \wedge_1$ | $[1, A]$ |
| 3. | Qa | $[2, D]$ |
| 4. | $P \wedge_2$ | $[1, A]$ |
| 5. | $Q\neg a$ | $[4, D]$ |
| 6. | Pa | $[5, A]$ |

This is a D -strategy. The same reasoning holds for $P\neg(a \wedge \neg a)$.

(4)

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|----|--|----------|
| 0. | $P((a \rightarrow a) \rightarrow b) \rightarrow b$ | |
| 1. | $Q(a \rightarrow a) \rightarrow b$ | $[0, A]$ |
| 2. | $P(a \rightarrow a)$ | $[1, A]$ |
| 3. | Qb | $[2, D]$ |
| 4. | Pb | $[1, D]$ |
| 5. | Qa | $[2, A]$ |
| 6. | Pa | $[5, D]$ |
- | | | |
|----|------|----------|
| 3. | Qa | $[2, A]$ |
| 4. | Pa | $[3, D]$ |
| 5. | Qb | $[2, D]$ |
| 6. | Pb | $[1, D]$ |

This is a D -strategy. If we omit positions 5 and 6, we still obtain an E -strategy.

(5a)

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|----|--|----------|
| 0. | $P((a \rightarrow b) \rightarrow a) \rightarrow a$ | |
| 1. | $Q(a \rightarrow b) \rightarrow a$ | $[0, A]$ |
| 2. | $P(a \rightarrow b)$ | $[1, A]$ |
| 3. | Qa | $[2, D]$ |
| 4. | Pa | $[1, D]$ |
- | | | |
|----|------|----------|
| 3. | Qa | $[2, A]$ |
|----|------|----------|

The left E -dialogue is won by P but not the right one. There is no strategy as long as $(D11)$ shall not be violated.

(5b)

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|----|--|----------|
| 0. | $P\neg\neg(((a \rightarrow b) \rightarrow a) \rightarrow a)$ | |
| 1. | $Q\neg(((a \rightarrow b) \rightarrow a) \rightarrow a)$ | $[0, A]$ |
| 2. | $P((a \rightarrow b) \rightarrow a) \rightarrow a)$ | $[1, A]$ |
| 3. | $Q(a \rightarrow b) \rightarrow a$ | $[2, A]$ |
| 4. | $P(a \rightarrow b)$ | $[3, A]$ |
- | | | |
|----|------|----------|
| 5. | Qa | $[4, D]$ |
| 6. | Pa | $[3, D]$ |
- | | | |
|----|--|----------|
| 5. | Qa | $[4, A]$ |
| 6. | $P((a \rightarrow b) \rightarrow a) \rightarrow a$ | $[1, A]$ |
| 7. | $Q(a \rightarrow b) \rightarrow a$ | $[6, A]$ |
| 8. | Pa | $[7, D]$ |

This an *E*-strategy and is easily extended to a *D*-strategy.

(6)

0.	$P((a \rightarrow b) \rightarrow (a \rightarrow c)) \rightarrow (a \rightarrow (b \rightarrow c))$		0		
1.	$Q(a \rightarrow b) \rightarrow (a \rightarrow c)$		$[0,A]$	1	
2.	$P(a \rightarrow (b \rightarrow c))$		$[1,D]$	0	
3.	Qa		$[2,A]$	1	
4.	$P(b \rightarrow c)$		$[3,D]$	0	
5.	Qb		$[4,A]$	1	
6.	$P(a \rightarrow b)$		$[1,A]$	2	
7.	Qa	$[6,A]$	3	7.	$Q(a \rightarrow c)$ $[6,D]$ 1
8.	Pb	$[7,D]$	2	8.	Pa $[7,A]$ 2
9.	$Q(a \rightarrow c)$	$[6,D]$	1	9.	Qc $[8,D]$ 1
10.	Pa	$[9,A]$	2	10.	Pc $[5,D]$ 0
11.	Qc	$[10,D]$	1	11.	Qa $[6,A]$ 3
12.	Pc	$[5,D]$	0	12.	Pb $[11,D]$ 2

This is again a *D*-strategy. If we omit positions 9–12 on the left branch and positions 11–12 on the right branch then we obtain an *E*-strategy. The numbers appearing to the right of the values of η are the *orders* of the respective assertions as they will be defined in Section 3.3.

(7) Let d_0 be the formula $f \wedge \neg f$ for some (atomic) f .

0.	$P(a \wedge ((b \rightarrow a) \rightarrow d_0)) \rightarrow c$		0		
1.	$Q(a \wedge ((b \rightarrow a) \rightarrow d_0))$		$[0,A]$	1	
2.	$P \wedge_1$		$[1,A]$		
3.	Qa		$[2,D]$	1	
4.	$P \wedge_2$		$[1,A]$		
5.	$Q(b \rightarrow a) \rightarrow d_0$		$[4,D]$	1	
6.	$P(b \rightarrow a)$		$[5,A]$	2	
7.	Qd_0	$[6,D]$	1	7.	Qb $[6,A]$ 3
				8.	Pa $[7,D]$ 2
				9.	Qd_0 $[6,D]$ 1

This can be completed so as to become a *D*-strategy; Qd_0 can be handled as in example (3), and on the left branch we then will have to add the steps occurring in the right branch as positions 7 and 8.

(8) Let d_0 be as in (7) and define recursively for $i = 0, 1, \dots$

$$\begin{aligned} e_i &= a \wedge ((b \rightarrow a) \rightarrow d_i) \rightarrow c, \\ d_{i+1} &= e_i \rightarrow d_0. \end{aligned}$$

Example (7) then gives a d -strategy for e_0 and, moreover, shows that there is a D -strategy for *each* e_{i+1} with Qd_i appearing in the positions 7 and 9 respectively. The D -strategy for e_i then contains assertions of orders up to $i+3$.

(9)

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|-----|--|----------|
| 0. | $P((a \rightarrow b) \rightarrow v) \rightarrow (c \rightarrow (b \rightarrow a))$ | |
| 1. | $Q(a \rightarrow b) \rightarrow v$ | $[0, A]$ |
| 2. | $P(c \rightarrow (b \rightarrow a))$ | $[1, D]$ |
| 3. | Qc | $[2, A]$ |
| 4. | $P(a \rightarrow b)$ | $[1, A]$ |
| 5. | Qa | $[4, A]$ |
| 6. | $P(b \rightarrow a)$ | $[3, D]$ |
| 7. | Qb | $[6, A]$ |
| 8. | Pa | $[7, D]$ |
| 9. | Qv | $[4, D]$ |
| 10. | Pb | $[5, D]$ |

The left-dialogue satisfies $(D10)$, $(D12)$, $(D13)$, and observing these rules Q has no possibility to continue it. The rule $(D11)$ is violated at place 6. If the formula v is chosen suitably then the dialogues can be extended so as to obtain a strategy, e.g., if v is $a \wedge b$ or $c \rightarrow a$ or $(a \rightarrow d) \rightarrow d$. However, for these choices of v already a D -strategy for the initial contention can be found.

(10)

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|----|---|----------|
| 0. | $P(\neg(a \rightarrow b)) \rightarrow (a \vee d)$ | |
| 1. | $Q\neg(a \rightarrow b)$ | $[0, A]$ |
| 2. | $P(a \vee d)$ | $[1, D]$ |
| 3. | $Q\vee$ | $[2, A]$ |
| 4. | $P(a \rightarrow b)$ | $[1, A]$ |
| 5. | Qa | $[4, A]$ |
| 6. | Pa | $[3, D]$ |
| 7. | Qc | $[4, D]$ |

Also this dialogue satisfies $(D10)$, $(D12)$, $(D13)$ and violates $(D11)$.

(11)

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|----|--|----------|
| 0. | $P(a \wedge ((\neg a \rightarrow b) \rightarrow c)) \rightarrow c$ | |
| 1. | $Q(a \wedge ((\neg a \rightarrow b) \rightarrow c))$ | $[0, A]$ |
| 2. | $P\wedge_1$ | $[1, A]$ |
| 3. | Qa | $[2, D]$ |
| 4. | $P\wedge_2$ | $[1, A]$ |
| 5. | $Q(\neg a \rightarrow b) \rightarrow c$ | $[4, D]$ |
| 6. | $P(\neg a \rightarrow b)$ | $[5, A]$ |

7.	$Q\neg a$	$[6,A]$	7.	Qc	$[6,D]$
8.	Pa	$[7,A]$	8.	Pc	$[1,D]$
9.	Qc	$[6,D]$	9.	$Q\neg a$	$[6,A]$
10.	Pc	$[1,D]$	10.	Pa	$[9,A]$

In the left dialogue, Q violates ($D11$) at place 9. If this place and the following place 10 are omitted, a D -strategy remains.

(12)

0.	$P(\neg(b \rightarrow (c \vee d))) \rightarrow (a \rightarrow b)$	
1.	$Q\neg(b \rightarrow (c \vee d))$	$[0,A]$
2.	$Pa \rightarrow b$	$[1,D]$
3.	Qa	$[2,A]$
4.	$P(b \rightarrow (c \vee d))$	$[1,A]$
5.	Qb	$[4,A]$
6.	$Pc \vee d$	$[5,D]$
7.	$Q\vee$	$[6,A]$
8.	Pb	$[3,D]$

P violates ($D11$) at place 8.

2 THE LITERATURE ON DIALOGUES

2.0

It was P. Lorenzen who, in addresses in 1958 and 1959, published as Lorenzen [1960; 1961], proposed the idea that an autonomous foundation of intuitionistic logic should be based on the concepts of a dialogue and of a strategy for dialogues. Emphasizing the autonomy of such a foundational approach, Lorenzen preferred to speak of a *constructive* or *effective* logic and avoided the more familiar name of *intuitionistic* logic. While the first descriptions of dialogues seemed to use only the properties named here ($D00$)–($D02$), it soon became clear that additional rules would be required if *only* intuitionistically provable formulas should be those which could be secured by strategies for dialogues. Such additional rules were formulated by Lorenz [1961] who defined (among other types) the kind of dialogues called D -dialogues here; they appear in Lorenzen [1962; 1967] and in Kamalah and Lorenzen [1967]. While these presentations attempted to arrive at an appropriate definition for a dialogue by specializing the general notion, a different approach was taken in [Lorenz, 1973] and [Lorenzen and Schwemmer, 1973] where there is considered at first a very narrow type of dialogue (permitting both P and Q to react only upon the immediately preceding step) which then is liberalised to types of dialogues which are, essentially, the E -dialogues of Section 1.

Lorenzen's basic idea is indeed a very attractive one. However, although described in a variety of articles and books, its presentations have been marred by a recurrence of ambiguous definitions and incomplete, if not erroneous proofs. And all these elaborations so far have suffered from two major defects.

2.1

The first defect is of a technical-mathematical character. If a new development of intuitionistic logic, based on the concepts of dialogues and strategies, shall be given, then one expects an equivalence theorem to be established which states that provability by strategies coincides with provability by one of the known calculi for intuitionistic logic. The proof of such a theorem remained missing for many years.

A first attempt to prove an equivalence theorem was made in Lorenz's dissertation [Lorenz, 1961]; it was repeated in [Lorenz, 1968]. A certain part of Lorenz's dissertation was corrected in [Stegmüller, 1964]; some claims made in other parts were proved while other ones were refuted in the Diplomarbeit of W. Kindt [1970]. Kindt's refutations were acknowledged in footnote 12 of [Lorenz, 1968] where it is said that a correction of the erroneous statements in [Lorenz, 1961] would require "ein paar detaillierte technische Vorbereitungen" (cf. also a similar remark in footnote No. 16); unfortunately, these few, detailed technical preparations have never been presented to the public, and the gaps in Lorenz's attempt still appear to be unfilled. (It is somewhat distressing that in the presumably authoritative collection of [Lorenzen and Lorenz, 1978] the article [Lorenz, 1968] is simply reprinted together with its footnotes; the part of [Lorenz, 1961] to which footnote No.12 refers has been omitted altogether.)

A first correct proof of an equivalence theorem was given by Kindt [1972]; however, the dialogues studied by Kindt are *not* D -dialogues but employ instead of ($D11$) a different rule. In [Lorenzen and Schwemmer, 1973, pp. 59 and 71] it is observed that the E -strategies (in the sense of Section 1) considered there give rise to a calculus of 'Dialogstellungen' which (at least in the propositional case) may be transformed into a calculus of Beth-tableaux such that provability by E -strategies implies intuitionistic provability. A new, and simpler approach to an equivalence theorem for D -dialogues was developed by Haas [1980] and it seems that the technical gaps contained in this work are only minor and can actually be filled. An attempt to prove an equivalence theorem for E -dialogues is contained in [Mayer, 1981] and Dr E. C. W. Krabbe informs me that an equivalence theorem for E -dialogues is contained in [Krabbe, 1982] and in [Barth and Krabbe, 1982].

The equivalence theorem stated in Section 1 was presented in [Felscher, 1981] and in a revised form in [Felscher, 1985].

2.2

The second defect from which the elaborations of Lorenzen's idea have suffered concerns the matter of foundations. The rules (D00)–(D01) are just precise descriptions of the intention that dialogues should proceed through applications of the argumentation forms. But when additional rules (the 'Dialog-Rahmenregeln' in the terminology of the Lorenzen school) had to be imposed, the question arose whether such rules could be explained as natural codifications of principles of argumentation. Such principles, if foundationally sound, would have to be based on an analysis of the use of dialogues as a means to establish a systematical, indisputable and convincing conduction of formal arguments.

When Lorenzen [1960; 1961] wrote about 'Sprachspiele', i.e., language games, he used this word in the sense of the ancients' $\alpha\gamma\omega\iota$, referring to a regulated (linguistic) process, the rules of which were to be governed by an insight which, although not further explained, was clearly assumed to be present. This attitude changed with [Lorenz, 1961] who, attempting a mathematical formalisation, began to make use of the concepts of a mathematical discipline known as the Theory of Games. Games there are mathematical objects, describing procedures as varied as whist and rummy at the one end and the games invented by warriors and economists at the other end. The rules then may be arbitrary: what matters is that they are adhered to; and the convention that a game is won because the other player can't draw any more may be brought about by rather odd rules of the game (such as, e.g., the categorical application of an equaliser). Matters were not improved when Lorenz [1961] observed that a change of dialogue rules would give rise to a type of dialogue the strategies for which would prove precisely the *classically* provable formulas. For this situation made it perfectly clear that the mathematical arbitrariness of the Theory of Games, being a tool to describe formally such different ways of reasoning as are classical logic and intuitionistic logic, could not possibly produce a philosophical foundation for either one of them. The mathematical apparatus for the Theory of Games was used heavily in the mathematical work of Kindt [1970; 1972]. On the other hand, phrases referring to 'dialogue games' have spread through a certain kind of literature where a mathematical terminology is borrowed in order to give at least the appearance of conceptual precision.

It appears that Lorenzen himself did not follow the fashion of a game-theoretical reduction. However the foundational discussions presented, e.g. in [Lorenzen and Schwemmer, 1973] are not based on an argumentative analysis; in particular, atomic statements and their negations are discussed with respect to a semantical distinction of true and false, and the difference between classical and intuitionistic logic is made to depend on the decidability ('Wahrheits-Definitheit': definiteness with respect to truth) of atomic statements. As crown's evidence for the generally unsatisfactory state of

a foundational discussion I may quote Kambartel [1979] who, in the very conclusion of this article, writes in respect to the problem of justification:

...Rechtfertigungsproblem. Dieses besteht darin, die *schematischen* Dialogspiele selbst von einem *argumentativen* Gebrauch der logischen Partikeln her zu begründen. Die dialogische Logik hat dieses Problem bisher dadurch überspielt, dass sie die Dialogspiele methodisch als ‘erste’ Festlegung des Gebrauchs der logischen Partikeln behandelt. Das dabei vernachlässigte, über die schematische Ebene hinausführende Rechtfertigungsproblem schlägt dann spätestens in der Rahmenregeldiskussion wieder durch. In der Tat werden dort ‘Rechtfertigungen’ für die Wahl solcher Regeln, z.B. neuerdings immament schematisch oder, wie zunächst geschehen, im eher intuitiven Rückgriff auf halb-schematisch analysierte Beispiele, beigebracht. Weder ‘technische’ Kriterien noch die Verallgemeinerung von Beispielen stellen aber bereits einen im engeren Sinne *normativen* Zugang zur Logik dar ...

2.3

Lorenzen’s argumentation forms have also been put to use by K. J. J. Hintikka, but this with quite different intentions. Hintikka, beginning with [Hintikka, 1968], developed what he calls a *game-theoretical semantics*, and a more recent series of articles on this topic have been collected in [Saari-nen, 1979]. A *semantical game* in Hintikka’s terminology may indeed be viewed as a dialogue in the sense of Section 1 although Hintikka restricts his attention to the single argumentation forms and nowhere cares to formulate game rules proper (such that the implied reference to mathematical games remains but an incantation). The point, however, is that Hintikka is concerned with a linguistical analysis of natural languages and *not* with a foundation of (classical or intuitionistic) logic. For this purpose, argumentation forms and dialogues are used as tools for the semantical evaluation of logically composite expressions (which may be more complex than first-order logic would permit to express) in domains governed by classical logic. There is, therefore, no connection of Hintikka’s work with that discussed in the present article.

3 FOUNDATIONS OF DIALOGUES

In this Section I shall develop an argumentative foundation for the use of particular types of dialogues, the *D*-dialogues, as a basis for intuitionistic logic. That such a foundation is wanted was outlined in subsection 2.2.

3.1 *The Argumentative Interpretation of Logical Operators*

There exists a well known provability interpretation of the logical operations (connectives and quantifiers) which then may be considered as being represented by Gentzen's calculi of proofs, i.e., natural deduction and the sequent calculus LJ; for details, cf. van Dalen's article in this volume. In the same manner, the argumentation forms may be viewed as expressing an *argumentative interpretation* of the logical operations, and as far as the positive connectives and the quantifiers are concerned this interpretation is obvious enough. Concerning implication, Y attacks $Xw_1 \rightarrow w_2$ by offering Yw_1 as an *admission* (or *local hypothesis*) and X may react by either answering with Xw_2 or attacking Yw_1 (provided w_1 is composite). Concerning negation, the situation is the same as in the case of the provability interpretation: if external, semantical references to truth and falsity shall be avoided, we must enrich the basic concept of provability by adding either refutability or absurdity as a primitive notion. Since we are aiming for intuitionistic logic, we introduce a constant Δ symbolising *absurdity* and then understand $\neg w$ as an abbreviation of $w \rightarrow \Delta$. The principle of *ex absurdo quodlibet* takes as its first form that he, X , who is forced to assert Δ then must concede, without further argument, any assertion made by Y . A speaker professing Δ thus brings himself into a position precluding any further debate, and so we may just as well omit this fatal step and state, as the second form of *ex absurdo quodlibet*, that Δ must not be asserted. This then explains why an attack Yw upon $Xw \rightarrow \Delta$, i.e., $X\neg w$, cannot be answered.

3.2 *Basic Principles for Dialogues*

Gentzen's calculi of proofs are easily explained in that they represent the weakest consequence relation for which the provability interpretation is valid. The connection between dialogues and the argumentative interpretation of logical operations is (not only more complicated but also) located on a different level: it is not the dialogues but the *strategies* for dialogues which will correspond to proofs. I thus formulate the *basic purpose* for the use of dialogues:

- (A₀) Logically provable assertions shall be those which, for *purely formal* reasons, can be upheld by a strategy covering every dialogue chosen by Q .

The dialogue rules (D00)–(D02), describing the use of argumentation forms, simply produce the (linguistic) material of the dialogue which then will have to be organised by the dialogue rules proper. Extending the intentions expressed in the formulation of the argumentation forms, I formulate the *argumentative intent* in the pursuit of a single dialogue:

- (A₁) A dialogue is, on the part of Q an attempt to put into doubt (to refute) the initial assertion made by P ; it is, on the part of P , an attempt to uphold this assertion, and if P succeeds in doing so this will mean that P wins the dialogue.

In the light of these intentions we now will have to clarify

- (b₀) how to determine the dialogue rules proper,
 (b₁) the notion that P wins a dialogue.

It must be emphasised that the concepts occurring here *cannot* be studied separately but must be analysed *simultaneously* and in constant regard of the purpose (A₀).

3.3 Dependence, Positive Dependence and Order

The notions to be discussed in this subsection are auxiliary. Let δ, η be a dialogue. I shall say that a statement *depends directly* on an earlier statement if it is either an attack upon or an answer to that statement; I define *dependence* to be the transitive and reflexive relation generated by direct dependence. Dependence, therefore, is an order relation, contained in the linear order given by δ ; since every statement, different from the initial one, depends directly on *exactly one* earlier statement, it follows that dependence defines the ordering of a *tree* on the set of all statements, i.e., on $\text{im}(\delta)$ with $\delta(0)$ as its top node. I define a *chain* to be a sequence of statements in $\text{im}(\delta)$ such that each of its members, except the first one, depends directly on its predecessor in that sequence; every chain κ thus arises from a path in the dependence tree by removing the nodes above $\kappa(0)$ from the path. Every chain is a subsequence of δ and δ itself is pieced together from various chains, some of which may only have one member.

While dependence is a relation defined between arbitrary statements, a second relation will be defined only between assertions of a dialogue δ, η . An assertion Xv is an *immediate positive dependent* of an earlier assertion Xw if it is an answer to an attack upon Xw ; I define *positive dependence* to be the transitive and reflexive relation generated by immediate positive dependence.

The relation of positive dependence leads to the following classification of assertions. The initial assertion and its positive dependence shall be *of order 0* ; if $Xv \rightarrow w$ or $X\neg v$ is *of order n* then an attack Yv shall be *of order n+1* , and so shall be the positive dependence of this attack. It follows that the P -assertions are exactly those of even order and the Q -assertions are exactly those of odd order.

3.4 Contentions and Hypotheses

When asserting the initial statement of a dialogue, P contends it to be defensible. In this sense, the initial assertion is *contended*, and in the same manner assertions of order 0 are contended. Assertions of order 1, if they do not arise as positive dependents of earlier ones, are attacks Qw_1 upon assertions $Pw_1 \rightarrow w_2$ or $P\neg w_1$; hence they are (global) *hypotheses* offered by Q , and then also their positive dependents are (particularisations of) hypotheses. Assertions of order 2, if they do not arise as positive dependents of earlier ones, are attacks Pw_2 upon hypotheses $Qw_1 \rightarrow w_2$ or $Q\neg w_1$; here P takes up the hypotheses by admitting Pw_1 as a (higher order) contention. Consequently, also these assertions are contended by P , and so are their positive dependents. Repeating this argument, it follows that *all* P -assertions are contended and that *all* Q -assertions are hypothetical. It will be advisable to observe the distinction made between global hypotheses in a dialogue as discussed here, and local hypotheses occurring as admissions in instances of argumentation forms.

Applying the argumentation forms, assertions of logically composite formulas are dissolved into assertions of lesser complexity: contentions are *upheld* and hypotheses are *developed*. Obviously, contentions $Pw_1 \wedge w_2$, $Pw_1 \vee w_2$, $P\forall xw$, $P\exists xw$ are upheld by holding up the immediate positive dependents (as chosen by P or prescribed by Q), and the same holds for the development of the analogous hypotheses. Consider now contentions $Pw_1 \rightarrow w_2$ where we include the case $P\neg w_1$ by writing it as $Pw_1 \rightarrow \Delta$; when attacked by Qw_1 then P may either uphold the answer Pw_2 or attack Qw_1 in order to force Q into a further development of this hypothesis. Similarly, if a hypothesis $Qw_2 \rightarrow w_2$ is attacked by Pw_1 then Q may develop it into the answer Qw_2 or Q may attack the contention Pw_1 .

The process of dissolving logically composite assertions comes to an end once atomic assertions have been reached. Atomic formulas asserted by Q are hypotheses, intended by Q as describing particular situations which serve to refute P 's contention to have a defensible initial assertion; they do *not* need further justification. Atomic formulas asserted by P , however, remain contentions *in need* of justification. Securing them by material insight, such as, e.g., illumination or revelation, is unacceptable (anyway and in particular) if *purely formal* defensibility has been claimed by P . There remains, therefore, only one possibility for P to assert an atomic formula Pa for purely formal reasons:

- (c_0) P may assert Pa only if Q admits Qa as a hypothesis *relevant* to the position of Pa .

For in that case P confronts Q with its own hypothesis to which Q cannot possibly object (in [Krabbe, 1982] this principle is mentioned with the ap-

propriate name of *ipse dixisti*). This principle (c_0) is purely descriptive in that it refers to a given, completed dialogue; it does not provide the means in order to enforce that, already during the performance of the dialogue, assertions Pa are made *only* in observance of (c_0). Making reference to the linear structure of a dialogue, we therefore strengthen (c_0) to

- (c_1) P may assert Pa only *after* Q has admitted Qa at an earlier, relevant position.

It appears that we now would have to clarify the notion of *relevance*, and a first attempt to do so would consist in producing a definition which describes, for every position of a dialogue, the set of hypotheses relevant for this position. I shall not proceed in this manner; rather, I shall introduce additional restrictive rules for dialogues with the effect that the family of *all* hypotheses occurring in a dialogue becomes *coherent* in the sense that its members, being admitted simultaneously, *do not* create distinctions of relevance: each of them is relevant for all atomic contentions asserted afterwards. The additional rules will, obviously, restrict the amount of information analysed in a dialogue. But the principal objects for us are strategies, not methods for winning a single dialogue, and no information will be lost if it only remains available within the system of dialogues belonging to a strategy.

3.5 How to Win a Dialogue

P wins a dialogue if it succeeds in holding up its contentions. During the course of a dialogue, the initial contention is dissolved into more and more specialised subcontentions, and this specialisation comes to an end with the contention of atomic formulas as regulated by (c_1). It is implicit in this conception that no composite contention is accepted as being upheld without further dissolution, for Q may always challenge it with an attack. Consider now a dialogue containing an atomic contention Pa which, for the moment, we assume as being of order 0. We then find a unique sequence of contentions, beginning with the initial contention Pv and ending with Pa , each of which (except the first one) is an immediate positive dependent of its predecessor. Consequently, Pa is the final step of a process by which Pv is narrowed down to more special contentions — and in view of (c_1) this final step may be asserted for purely formal reasons. Of course, other sequences of specialisations of Pv , ending with other atomic formulas, may be possible, but since we are considering provability by strategies, these other possibilities are covered by other dialogues which a strategy will have to take into account. We thus arrive at

- (WA) The initial contention is considered as having been upheld successfully if P has asserted an atomic contention of order 0.

Obviously, the same reasoning can be used to say that a contention of order n can be considered as having been upheld successfully if P has narrowed it down to an atomic contention of order n — but this observation remains without consequences.

There is, however, another way to consider the initial contention as having been upheld successfully; it rests on the principle of *ex absurdo quodlibet* applied to dialogues (and not only to argumentation forms). As a simplest situation, consider a hypothesis Qw_1 of order 1, arising as an attack upon $Pw_1 \rightarrow w_2$ of order 0, and assume that, after a sequence of positive dependents of Qw_1 , the only possibility left to Q would be the assertion of an absurdity $Q\Delta$. In that case, the hypothesis Qw_1 has been developed (and we may assume: by P 's prodding) into an absurdity and, therefore, has itself been shown as untenable. But as this hypothesis had been granted in the attack Qw_1 , we conclude by *ex absurdo quodlibet* that $Pw_1 \rightarrow w_2$ can be upheld *without* any further argument. More generally, a hypothesis Qr_1 of order $n+1$, $n > 0$, arising as an attack upon $r_1 \rightarrow r_2$ of order n , is itself a development of the earlier hypothesis $Qs_1 \rightarrow s_2$ of order $n-1$ which gave rise to an attack Ps_1 of which $Pr_1 \rightarrow r_2$ is a positive dependent. Consequently, if Qr_1 can be shown as leading to an absurdity, then also $Qs_1 \rightarrow s_2$ must be considered as leading to an absurdity. Descending from $n-1$ to 1, we conclude that already the first hypothesis Qw_1 of order 1, giving rise to the higher order hypotheses resulting in $Qs_1 \rightarrow s_2$, leads to an absurdity, and thus again $Pw_1 \rightarrow w_2$ can be upheld without further argument. We thus arrive at

(WB) The initial contention is considered as having been upheld successfully if (P has not asserted an atomic contention of order 0 but) Q has been brought into a position where its only possibility to continue would be the assertion of an absurdity.

Of course, at the present stage of our discussion no reason is visible why Q may become so restricted in its possibilities as is supposed in (WB); this will become clear after the following sections. What *can* be said already here is that if a dialogue is won by P then its last position is even (i.e., the last move is P 's), and if the initial contention does not contain a negation then the dialogue can be won only according to (WA).

3.6 Ramifications

I shall now discuss how to avoid the distinctions of relevance as they appear in condition (c_0). Let us begin by considering more closely the situation described there: let δ, η be a dialogue with positions j, k such that $j < k$, $\delta(j) = Qa$, $\delta(k) = Pa$ where a is atomic. Since both these assertions depend on $\delta(0)$, there exist chains κ_0, κ_1 from $\delta(0)$ to $\delta(j)$ and to $\delta(k)$, and it follows from $j < k$ that κ_1 cannot be an initial part of κ_0 . Consequently,

either κ_0 is an initial part of κ_1 or κ_0, κ_1 ramify at a position preceding $\delta(j)$. The first case is unproblematical. For let n be the last argument of κ_0 such that $\kappa_0(n) = \delta(j)$; being an atomic hypothesis, $\kappa_0(n)$ must be an attack upon $\kappa_0(n-1) = Pa \rightarrow w$ which then is answered by $\kappa_1(n+1) = Pw$, for this is the only way in which the atomic hypothesis Qa can have dependents in κ_1 . Thus Pa is Pw or a dependent of Pw , and Qa must certainly be considered as relevant for Pa . In the second case, however, the ramification may cause distinctions of relevance.

I shall say that there is a *ramification at n* for two chains κ_0, κ_2 of a dialogue if both chains coincide for all i such that $i \leq n$, if further both chains are defined (at least) for $n+1$ and if $\kappa_0(n+1)$ is different from $\kappa_1(n+1)$, i.e., if $\kappa_0(n+1), \kappa_1(n+1)$ are stated at different positions of the dialogue. A ramification then arises in one of the three following ways:

- (1) $\kappa_0(n+1), \kappa_1(n+1)$ are different attacks upon $\kappa_0(n)$;
- (2) $\kappa_0(n+1), \kappa_1(n+1)$ are different answers to the attack $\kappa_0(n)$ upon $\kappa_0(n-1)$;
- (3) $\kappa_0(n)$ is an attack Yw_1 upon $\kappa_0(n-1) = Xw_1 \rightarrow w_2$, $\kappa_i(n+1)$ is an attack upon $\kappa_0(n)$ and $\kappa_{1-i}(n+1)$ is the answer Xw_2 to $\kappa_0(n)$, $i = 0, 1$.

The attacks in (1) shall be called *distinct* if not only their positions but also the statements made by them are different; otherwise they are only *repeated attacks*. In the same way, I shall speak of *distinct* and of *repeated* answers. In this subsection I shall be concerned with ramifications of the first two types.

Distinct *attacks* are possible upon assertions $Xw_1 \wedge w_2, X\forall xw$. If P contends, say, $Pw_1 \wedge w_2$ then P certainly should be able to contend both Pw_1 and Pw_2 . But a hypothesis arising during the analysis of Pw_1 (e.g., if w_1 is $a \rightarrow a$) will *not* be relevant during the analysis of Pw_2 (e.g., if w_2 is $b \rightarrow a$) and vice versa: such hypotheses are admitted for *one* but not for the *other* subcontention. Distinct attacks upon contentions, therefore, do cause distinctions of relevance. In a strategy, however, the possibility of distinct attacks by Q is already taken into consideration in that it leads to different dialogues which all have to be won by P . Consequently, there will be no loss of information if such distinct attacks are excluded from every single dialogue. On the other hand, a hypothesis $Qw_1 \wedge w_2$ is present already *before* any ramification caused by different attacks and it should remain in effect with its *complete content* also after the ramification. If, for instance, during the development of Qw_1 the moves of P lead Q to admit Qa then this hypothesis should be considered as coherent with Qw_1 as well as with $Qw_1 \wedge w_2$ and also with Qw_2 : Qa should be relevant also for any development of Qw_2 .

Distinct *answers* can be given to attacks upon assertions $Xw_1 \vee w_2, X\exists xw$. If P contends, say, $Pw_1 \vee w_2$ then it will have to contend only *one* of

Pw_1, Pw_2 . Again now, hypotheses arising during the analysis of one of these answers (e.g., if w_1 is $\neg a$) will not be relevant during the analysis of the other (e.g., if w_2 is a). Distinct answers to attacks upon contentions, therefore, do cause distinctions of relevance. However, if P has answered an attack with, say, Pw_1 then a second, later answer with Pw_2 would be useful only if Pw_1 could not be upheld successfully and if P now would try a second attempt. But in a strategy we can demand that P knows what it is doing and does not proceed by trial and error, and there will be no loss of information if we exclude distinct answers to attacks upon contentions. On the other hand, if Q were to answer an attack upon, say $Qw_1 \vee w_2$ with both Qw_1 and Qw_2 then it would grant not only the content of $Qw_1 \vee w_2$ but actually that of $Qw_1 \wedge w_2$. In a strategy, however, P has to provide dialogues for all possible answers, and there will be no loss of information if we also exclude distinct answers to attacks upon hypotheses.

In order to discuss repeated attacks and repeated answers, we may now assume that the two chains κ_0, κ_1 ramifying at n , are chosen as being maximal, i.e., as branches of the dependence tree. I now define inductively the notion of *corresponding couples*: $\kappa_0(0), \kappa_1(0)$ form a corresponding couple; if $\kappa_0(i), \kappa_1(i)$ form a corresponding couple then $\kappa_0(i+1), \kappa_1(i+1)$ shall form a corresponding couple if they either appear at the same position of the dialogue or if they are (at least) identical as signed expressions *and also* are identical in their *mode* within the dialogue, i.e., as attacks or answers referring to $\kappa_0(i), \kappa_1(i)$. Let us assume now that the ramification at n arises under repetitions. Then $\kappa_0(n+1), \kappa_1(n+1)$ still form a corresponding couple, and we may look for all corresponding couples $\kappa_0(n+i), \kappa_1(n+i)$. If these couples exhaust already one of the two chains, say κ_0 , then that part of κ_0 which starts at $n+1$ does not contain any information which is not available in the corresponding part of κ_1 . No information will be lost if that part of κ_0 (or the corresponding part of κ_1) is omitted from the dialogue, and so we may exclude the repetition at $n+1$ which gave rise to the twofold presence of that part.

It remains to consider the case that there exists a common argument m of κ_0, κ_1 , $n+1 < m$, such that $\kappa_0(m), \kappa_1(m)$ is not a corresponding couple. Let m be minimal for this property and observe that either P or Q acts at m in both of κ_0, κ_1 . If P attacks at m , say $Qw_1 \wedge w_2$ with Pw_1 in the one and with Pw_2 in the other chain, then these distinct attacks could have been carried out *without* the repetitive ramification at n (and with an acceptable ramification at the position of $Qw_1 \wedge w_2$ instead). If P answers at m , say Qv with Pw_1 and Pw_2 , then the repetitive ramification at n just hides the fact that there is actually a ramification caused by distinct answers. The same types of ramifications occur if Q in both κ_0, κ_1 attacks m or answers at m . Finally, if either P or Q attacks m in the one and answers m in the other chain then the repetitive ramification at n delays an actual ramification of type (3). Again now, *either* this ramification of type (3) could have been

carried out already at n or there are restrictions on the execution of such ramifications (as they actually will be discussed in the next section) and the repetitive ramification at n is employed in order to circumvent these restrictions.

Consequently, in *all* cases repetitive ramifications either can be avoided immediately or lead to distinctions of relevance of the sort discussed already for distinct ramifications. In any case, therefore, they may be excluded without loss of information.

The conditions effecting the exclusion of different answers and of different attacks upon contentions are precisely (D12) and (D13).

3.7 Nested Attacks and Nested Answers

There are good reasons why, in a dialogue, a certain answer upon an attack by Q is not stated immediately but only after some delay. The necessity to observe (c_1) will cause such situations if

a contention $Pw_1 \rightarrow w_2$ has been attacked and the admitted hypothesis Qw_1 needs further elaboration in order to permit P either to assert Pw_2 or to force Q into an absurdity,

or a contention, itself stated already under a hypothesis, has been attacked and now the earlier hypothesis needs further elaboration.

Delayed answers, therefore, cannot be excluded. It is such delays which cause a nesting of attacks and, thereby, a nesting of answers.

Let δ, η be a dialogue and let $\delta(m), \delta(n)$ be assertions such that $m-n \equiv 0 \pmod{2}$, let $\delta(i)$ be an attack upon $\delta(m)$ and let $\delta(j)$ be an attack upon $\delta(n)$ (whence also $i-j \equiv 0 \pmod{2}$). I shall call $\delta(j), \delta(i)$ a pair of *nested attacks* if $j < i$ and if the attack $\delta(j)$ is still open at i ; in that case $\delta(i)$ is the *inner* attack and $\delta(j)$ is the *outer* attack. I shall speak of *H-nested attacks* if the inner attack is a hypothesis; in that case, both m, n are even, $\delta(m)$ is of the form $Pw_1 \rightarrow w_2$ or $P\neg w_1$ and $\delta(i)$ is Qw_1 .

Consider a pair of *H-nested attacks* as above and let Ps be the contention which P would have to state in order to answer the outer attack. The question then arises whether the hypothesis Qw_1 can be considered as relevant for Ps and its dependents, since P contended the assertion $\delta(n)$ *before* Qw_1 has been admitted. For instance, if $\delta(n)$ is $Pr \rightarrow s$ then, by contending it, P states that it is prepared to hold up Ps under the hypothesis Qr — an additional hypothesis such as Qw_1 is, at this stage, not available. Obviously, hypotheses available at the position n may be used as well, and we also can assume that distinctions of relevance have already been excluded at least as far as the positions n, m . The complete content now of $\delta(m)$, i.e., of $Pw_1 \rightarrow w_2$ or $P\neg w_1$ is, again, that P is prepared to hold up Pw_2

(or, for that matter, even $P\Delta$) under the hypothesis Qw_1 . Therefore, the hypothesis Qw_1 should be counted as being relevant if P actually fulfils the obligation to assert PW_2 (or $P\Delta \dots$): Qw_1 *together with* Pw_2 preserves the complete content of $\delta(m)$. We thus arrive at the principle

- (d_0) For every pair of H -nested attacks: the hypothesis granted by the inner attack may be used as relevant in order to contend the answer to the outer attack *provided* P also contends the answer to the inner attack.

The dialogue won by P in example (9) satisfies (d_0) for the H -nested attacks with $n = 1$, $m = 4$, $j = 3$, $i = 5$; the hypothesis $Qw_1 = Qa$ is used in order to contend the answer $\delta(6)$ to $\delta(j)$ at 8, and conversely also this answer itself is employed in order to develop the hypothesis Qb which is needed when contending the answer $\delta(10)$ to $\delta(i)$.

The principle (d_0) is purely descriptive; it does not provide the means in order to enforce that only the situation described as desirable occurs. If the inner attack is upon $P\neg w_1$ then P will never be able to fulfil the obligation expressed in (d_0), and if we wish to avoid an additional label declaring Qw_1 to be irrelevant for Ps then another formulation is wanted. But also if $\delta(m)$ is $Pw_1 \rightarrow w_2$ with $w_2 \neq \Delta$, a different formulation would be useful: the example (10) shows a pair of H -nested attacks violating (d_0), but P has won the dialogue in accordance with (WA) and Q , if it is left to respect (D12), (D13), cannot continue: there just is no position left for P to state the answer to the inner attack. We thus formulate a *rule* which forces P to contend this answer *before* it makes use of the hypothesis:

- (d_1) For every pair of H -nested attacks: the inner attack must have been answered before the outer attack may be answered.

In this manner, we now have also a well-defined nesting of answers.

Consider now a pair of nested attacks upon contentions which is not H -nested; this means that the inner attack is symbolic. The statement of such an attack does not create any hypotheses possibly needed for the contention of the answer Ps to the outer attack. Still, there may be various reasons for P *not* to answer the outer attack $\delta(j)$ immediately at $j+1$, but to delay this answer to a position following that of the inner attack: actually the observance of (d_1), together with that of (c_1) may be one such reason. The example (12) communicated to me by Dr. E. C. W. Krabbe, shows a dialogue satisfying (D10), (D12), (D13) and (d_1), but the outer attack at 3 is answered at 8 while the inner attack at 7 remains open. Given the outer attack 3, also the hypothesis $\delta(5)$ arises as an inner attack, and observance of (d_1) forces P to state the contention $\delta(6)$ before $\delta(5)$ can be used in order to state the answer Pb to the outer attack. In avoiding the answer to the inner attack at 7, P now *fails* to uphold the contention $\delta(6)$. Thus (d_1) has

been observed, but the promise to uphold $\delta(6)$ is given lip service only. It follows from this example that, in order to observe the full meaning of (d_0) we have to strengthen the rule (d_1) to

(d_2) For every pair of nested attacks upon contentions: the inner attack must have been answered before the outer attack may be answered.

As for nested attacks upon hypotheses, it follows from the definition of a strategy that, if P has a strategy at all, then P *a fortiori* has a strategy respecting

(d'_2) For every pair of nested attacks upon hypotheses: the inner attack must have been answered before the outer attack may be answered.

There is reason to conjecture that also the converse implication holds, and a proof should be related to that of the Extension Lemma mentioned in Section 1. The idea of such a proof is illustrated by example (11): violating $(D11)$, Q may try to withhold a certain hypothesis (e.g., absurdity), but P knows from the strategy how the answer to the outer attack had to be treated if it was stated immediately after this attack.

3.8 *D-Dialogues*

It was the purpose of the last two subsections to look for restrictions on dialogues which would permit us to avoid distinctions of relevance. We thus arrived at the rules $(D12)$, $(D13)$ in subsection 3.6 and at rule $(D11)$ which is the conjunction of (d_1) and (d'_2) in subsection 3.7. Having imposed these rules, let us look once more at the family of hypotheses occurring in a dialogue. Clearly, every positive dependent of a hypothesis Qw is nothing but a specification or an instantiation of Qw ; and dependents of higher order, coming from intermediary contents, keep this character as well. Passing through a chain of dependents, we see that the hypotheses occurring there form a coherent set, i.e., a set of simultaneously admitted assumptions without distinctions of relevance. Different chains of dependents are joined together with the help of delayed attacks and answers, and the three types of ramifications which thus may arise were described at the beginning of subsection 3.6. The presence of our rules now insures that all those ramifications are excluded which would cause distinctions of relevance. What remains permitted are ramifications of type (3) with properly nested answers and ramifications of type (1) caused by different attacks upon a hypothesis. If P attacks a hypothesis Qw a first time, it forces from Q a certain system of specifications of the hypothesis Qw asserted by Q in the beginning; if P attacks Qw a second time then it may obtain a different system of specifications which, nevertheless, still is a system of specifications of this *same* hypothesis Qw : once Q has admitted Qw then it must

bear the consequences of being forced into specifications, and the specified hypotheses brought forward under the different answers given by Q to the *same* attack of P upon Qw express, when considered simultaneously, *more* than is contained in Qw alone, and thus they *do* create distinctions of relevance. It must be emphasised that this difference in the effects of attacks and answers is fundamental.

We thus find that, in the presence of $(D11)$, $(D12)$, $(D13)$, the set of *all* hypotheses occurring in a dialogue is coherent in the sense that no distinctions of relevance appear. Consequently, the condition (c_1) becomes $(D10)$, and our dialogues are D -dialogues.

At this point, a methodological observation appears to be appropriate. In the preceding two subsections I have presented arguments resulting in the introduction of additional rules with the purpose to avoid distinctions of relevance $((D12), (D13), (d_0))$. In subsection 3.6 the exclusion of certain moves in a dialogue was supported with the observation that no information will be lost if the possibilities, excluded from single dialogues, remain present in strategies; in subsection 3.7 the introduction of (d_0) was explained with the necessity to preserve the complete content. It should be noticed very clearly that these argumentations are based on an *informal understanding of purposes*; they are *not* justifications based on mathematical theorems. As a matter of fact, as long as we abstain from a formal definition of relevance, we cannot even formulate a theorem saying that strategies for dialogues *with* precautions on relevance (?) prove the same formulas as do strategies for dialogues with $(D12)$, $(D13)$, (d_0) .

3.9 How to Win a D -dialogue

I have defined in subsection 3.5 what it means that P wins a dialogue. On the other hand, there is the purely formalist definition, taken from the literature and mentioned in Section 1, that a D -dialogue is won by P if Q has no way to continue. It remains to be shown that both notions are equivalent with respect to strategies.

If a D -dialogue has been won according to (WB) then it also has been won in the formal sense. If a D -dialogue has been won according to (WA) then it contains an atomic contention Pa of order 0 which, therefore, must be an answer. Consequently, Q cannot continue the dialogue at this position by referring to Pa . We now can show:

If for some formula v , P has a strategy to win with (WA) , (WB) all D -dialogues for v then P also has a strategy to win in the formal sense all D -dialogues for v . For consider a strategy the branches of which are won with (WA) , (WB) . These branches could not be continued by Q if Q would have to respect the rules for E -dialogues. We thus obtain an E -strategy if, where necessary, we cut off ends of branches when Q begins to violate the

rule (*E*). It then follows from the Extension Lemma that this *E*-strategy may be extended again to a *D*-strategy.

For the converse implication, it can even be shown that a *D*-dialogue δ, η which is won in the formal sense is won also according to (*WA*), (*WB*). As this is clear if, during the dialogue, *P* has stated an atomic contention of order 0, we may assume now that *P* has *not* stated an atomic contention of order 0; (*WB*) then will hold if we can prove that *Q* could continue if it were to state an absurdity: there still should exist a (last) open attack of the form $P \neg w$. Before continuing the proof, it will be useful to prepare some auxiliary notions.

Let δ, η be a dialogue; let n be an odd position and assume that *Q* has already stated $\delta(n)$. Then $\xi_0(n)$ shall be the number of contentions which, after this statement, still may be attacked by *Q* at later positions, and $\xi_1(n)$ shall be the number of contentions which, being attacks, still may be answered by *Q* at later positions — and here I expressly *include* the unspeakable answers $Q\Delta$ to attacks Pw upon some $Q \neg w$. The number $\xi(n) = \xi_0(n) + \xi_1(n)$ is called the *degree* (of freedom) of n . The following characterisation will be useful:

$\xi(n)$ is the difference $a(n) - d(n)$ where $a(n)$ is the number of attacks upon hypotheses $Qw_1 \rightarrow w_2$, w_1 not atomic, which are contended before n , and $d(n)$ is the number of atomic *P*-answers contended before n . This follows from the following observations in which $2i+1$ is a position of δ, η . If $\delta(2i)$ is an attack then $\xi(2i-1) \leq \xi(2i+1) \leq \xi(2i-1) + 1$, and the right inequality becomes equality if, and only if, $\delta(2i)$ is an attack upon a hypothesis $Qw_1 \rightarrow w_2$ such that w_1 is not atomic (whereas w_2 may be Δ). If $\delta(2i)$ is an answer which is not atomic then $\xi(2i-1) = \xi(2i+1)$. If $\delta(2i)$ is an answer which is atomic then $\xi(2i-1)$ must be positive (for otherwise *Q* could not act at $2(i+1)$), and $\xi(2i+1) = \xi(2i-1) - 1$.

Consider now a *D*-dialogue. It follows from (*D12*) (*D13*) that every assertion has *at most one* atomic immediate positive dependent; consequently, every assertion has *at most one* atomic positive dependent. Let now Pa be an atomic contention of *positive* order; there then exists a first (highest) contention Pw_a of which Pa is a positive dependent, and Pa, Pw_a have the same order. By the preceding remark, Pw_a is unique, and as it has no positive predecessor, it must be an attack upon a hypothesis $Qw_a \rightarrow u_a$ (where u_a may be Δ).

I now resume the proof where it was interrupted. Let $2i$ be the last position of the dialogue. Then $\delta(2i)$ cannot be a composite formula (since that could be attacked by *Q*) nor can it be a symbolic attack (since that could be answered); thus it must be an atomic contention Pa , necessarily of positive order. Let Pw_a be the unique, highest contention determined by Pa as above. If Pw_a is Pa then u_a must be Δ (for otherwise *Q* could answer), and thus (*WB*) holds. Assume now that Pw_a is different from Pa ; then w_a is not atomic. It now will be sufficient to show that $\xi(wi-1)$ is positive

— because the possibilities numbered by $\xi(i-1)$ cannot comprise attacks or answers which actually could be stated (for in that case Q could still continue), and thus there must be at least one open attack to be answered only by absurdity. I thus have to prove that the difference $a(2i-1)-d(2i-1)$ is positive. Let Pb be an atomic answer which contributes to $d-(2i-1)$; it is of positive order and thus determines the unique attack Pw_b . Since Pb is an answer and Pw_b is an attack, Pb must be a proper dependent of Pw_b . Thus w_b is not atomic and, therefore, the attack Pw_b contributes to $a(2i-1)$. It follows that the map φ sending b into w_b is an injection of the set of contentions contributing to $d(2i-1)$ into the set of contentions contributing to $a(2i-1)$. Since w_a is not atomic, the latter set contains Pw_a ; the former set, however, does not contain Pa . Consequently, $a(w_i-1)$ is strictly larger than $d(2i-1)$.

3.10 Intuitionistic versus Classical Logic

As was mentioned in Section 2, Lorenz [1961] has observed that a change in the rules for D -dialogues produces a class of dialogues which I shall call *C-dialogues*, such that the formulas provable by C -strategies are precisely the classical provable formulas. **The change leading from D -dialogues to C -dialogues consists in**

cancelling (D11) and (D12) for P , but leaving them in effect for Q .

(If I understand Lorenz's and Lorenzen's writings correctly then they seem to demand the cancellation for P of (D12) only; the examples (2b) and (5a) show that this would not suffice.) It is not hard to see that C -strategies prove *only* classically provable formulas. For the case of *propositional* logic, the converse implication (i.e., every classically provable formula can be proved by a C -strategy) can be seen as follows. Observe first that an intuitionistically provable formula, being provable by a D -strategy, is trivially provable by a C -strategy. It is well known that if w is a classically provable formula then $\neg\neg w$ is intuitionistically provable; assume now that w is not intuitionistically provable. Every D -dialogue δ, η for $\neg\neg w$, won by P , begins with attacks $\delta(1) = Q\neg w, \delta(2) = Pw$, and since we assume that the part beginning at position 2 is *not* a D -dialogue for w won by P , there must be positions below 2 at which P attacks $\delta(1)$ again. If we compare the branches in the dependence tree and look for the first positions at which they differ, we will find, refining the discussion in subsection 3.6, that this happens at contentions which could be obtained *without* the repetitive first part of the branch if repeated answers or answers in disregard of (D11) were permitted to P . Permitting such answers in C -dialogues, it can be shown that a D -strategy for $\neg\neg w$ can be rebuilt into a C -strategy for w .

The mathematical fact that *C*-strategies can be used for classical logic is, in principle, not surprising; other proof-theoretical systems, e.g., those of Hilbert-type, can also be used for many varieties of logics. Rather than leading to amazement over the universal applicability of a mathematical tool (trees and strategies), this situation should teach us to emphasise the fundamental differences between intuitionistic and classical logic.

For the provability interpretation, as represented by Gentzen's calculi, Curry [1963, p. 260] has attempted a provability *explanation* of classical negation with help of his concept of *complete absurdity*, but this hardly will be considered to be a conceptual *foundation*. For the argumentative approach presented here, classical logic *cannot* be given a foundation by simply changing formal details of a foundation for intuitionistic logic. If we want to explain the rules governing classical negation then there appears to be no way to avoid the semantical notions of *true* and *false*: without these notions we cannot explain why distinctions of relevance may be discarded as it is done when *P* is permitted to repeat answers and to disregard (*D11*). Thus, for classical logic, the entire conceptual frame employed for the foundation of intuitionistic strategies, has to be abandoned: there is no use for contentions and hypotheses, for defendability by purely formal reasons and for considerations of relevance. What *is* required, is a completely different conceptual framework, based on the notions of true and false and on the distribution of truth-values under logical operations. The foundation of classical logic within such a framework is well known, and the elegant formulation of classical tableaux due to Smullyan [1968] may easily be read as to depict a dialogue-strategy leading to a failure of the attempt to falsify a formula. Again, the argumentative explanation of winning a dialogue according to (*A*), (*WB*) is only formally related to the closure of branches in Smullyan's tableaux which *always* means the advent of absurdity.

4 APPENDIX: CONCEPTS CONNECTED WITH THE EQUIVALENCE THEOREM

The equivalence theorem, formulated in Section 1, states the existence of certain transformations between strategies and proofs in the calculus LJ; the proof of this theorem cannot be presented here. It may, however, be instructive for the reader to become familiar with some concepts which originally were developed for this proof. For details which have to be suppressed here I refer to Felscher [1981; 1985].

The reader will have noticed that among the examples, listed at the end of Section 1, there is none which treats a formula with quantifiers. But this is no serious omission since the argumentation forms for quantifiers are, so to speak, the direct generalisations of the forms for conjunction and disjunction to the infinite case. For strategies, however, this treatment of quantifiers has

the effect that there may occur infinite ramifications: if S, δ, η is a strategy and if a node e carries as $\delta(e)$ either a formula $P\forall xw$ or an attack $P\exists$ upon a formula $Q\exists xw$ then the tree S has an infinite ramification at e — every term t determines a lower neighbour of e , carrying either an attack Qt or an answer $Qw(t)$. Although the branches of S must be finite (as follows from (S0)), the strategy itself is an infinite object. It is obvious that this is a clear disadvantage of strategies as compared to the more usual notions of proof. I now shall abstract a *finite* object from a strategy, its skeleton.

Let H be a class of dialogues as in Section 1. An H -skeleton for a formula v is a triplet S, δ, η with the same properties as an H -strategy for v *except* that in (S1) certain nodes e are excepted and, instead, are covered by

- (S1_e) If $\delta(e)$ is $P\forall xw$ then only one lower neighbour of e carries an attack upon $\delta(e)$, and this attack is Qy where y is a variable not occurring free in any *expression* $\delta(h)$ with $h \geq e$. If $\delta(e)$ is an attack $P\exists$ upon $Q\exists xw$ then only one lower neighbour of e carries an answer, and this answer is $Qw(y)$ where y is a variable not occurring free in any *expression* $\delta(h)$ with $h \geq e$.

As is usual, the variable y will be called the *eigenvariable* in these situations. It is clear that every H -strategy contains various H -skeletons, and it is not hard to see that, conversely, every H -skeleton can be extended to an H -strategy. This observation has the important consequence that it suffices to consider H -skeletons which, having finite trees, are more easily handled in induction proofs. For instance, the Extension Lemma of Section 1 is proved in the form that every E -skeleton can be extended to a D -skeleton.

Unfortunately, E -skeletons still have certain undesirable properties. Consider the example of a formula $\exists xa \rightarrow \exists xa$ where a is atomic; there are two E -strategies, viz.

0.	$P\exists xa \rightarrow \exists xa$		0.	$P\exists xa \rightarrow \exists xa$	
1.	$Q\exists xa$	[0,Q]	1.	$Q\exists xa$	[0,A]
2.	$P\exists$	[1,A]	2.	$P\exists xa$	[1,D]
3.	$Qa(y)$	[2,D]	3.	$Q\exists$	[2,A]
4.	$P\exists xa$	[1,D]	4.	$P\exists$	[1,A]
5.	$Q\exists$	[4,A]	5.	$Qa(y)$	[4,D]
6.	$Pa(y)$	[5,D]	6.	$Pa(y)$	[3,D]

In the right skeleton, the attack at 3 is answered with the substitution term y at 6; this answer must be delayed because the choice of the substitution term depends on the eigenvariable y appearing at 5. There are no phenomena of an analogous type in, say, the sequent calculus; in Lorenzen and Schwemmer [1973] and in Haas [1980], where an informal use of E -skeletons is made, the possibility that this situation might occur has been overlooked.

In order to circumvene this difficulty, I introduce the concepts of a *formal dialogue* and of a *formal strategy*, making use of the *formal argumentation forms* for $\forall$ and $\exists$:

$Q\forall$:	assertion: $Q\forall xw$	$P\forall$:	assertion: $P\forall xw$
	attack: Pt		attack: Qy (<i>eigenvariable</i>)
	answer: $Qw(t)$		answer: $Pw(y)$
$Q\exists$:	assertion: $Q\exists xw$	$P\exists$:	assertion: $P\exists xw$
	attack: $P\exists$		attack: Qt
	answer: $Qw(y)$ (<i>eigenvariable</i>)		answer: $Pw(t)$.

I define a *formal E-dialogue* in exactly the same way in which I defined an *E-dialogue*, only now in (D01), (D02) the formal argumentation forms are used for quantifiers *and* the eigenvariable condition is imposed at the position indicated. The adjective *formal* then refers to the fact that, contrary to the intuitive understanding, in the attack Qt the term t is stated already by Q ; eigenvariables chosen at a later position then *must* respect these expressions Qt . I define a *formal E-strategy* in the same way in which I defined an *E-strategy*, but now with formal dialogues instead *and* with the following changes: there is *only one* possibility for Q taken into account for

answering an attack $P\exists$	(case $Q\exists$),
making an attack Qy	(case $P\forall$),
making an attack Qt	(case $P\exists$).

It then is obvious that every formal *E-strategy* can be transformed into an *E-skeleton*; it can be shown that, conversely, every *E-skeleton* can be transformed into a formal *E-strategy*.

It is the formal *E-strategies* which can be set into correspondence with LJ-proofs.

It follows from these observations that the disadvantage of dialogues consisting in

1. the treatment of quantifiers as infinite conjunctions and disjunctions disregarding Frege's discovery of finitary quantifier rules made possible by the use of free variables, and
2. the ensuing appearance of infinite strategies

is only apparent. It arose because we wanted to use the *same* argumentation forms (concerning quantifiers) for *both* P and Q ; it could have been avoided if, from the outset, we would have studied strategies instead of dialogues. This illustrates once more the difficulty, mentioned at the beginning of Section 3.2, that it is not the dialogues but the strategies which correspond to proofs: working with dialogues, we have to describe *in advance* the branches

of strategies which themselves are to be defined only with the help of these dialogues.

It also should be observed that, in contrast to Gentzen's calculi for provability, strategies and dialogues do *not* appear as natural representations of the relation of *provability from hypotheses* but only as those of the relation of *absolute provability*. Of course, if M is a finite set of sentences and m is a conjunction of these sentences then, for every sentence w , the sequents

$$M \Rightarrow w \quad \text{and} \quad \Rightarrow m \rightarrow w$$

are simultaneously derivable in LJ, and this permits us to reduce the provability of w from the hypotheses M to the absolute provability of $m \rightarrow w$. It also is obvious that a dialogue, discussing the derivability of $M \Rightarrow w$, should begin with an initial list of the Q -formulas determined by M , followed (or preceded) by the P -formula Pw . But no general rule on how to proceed from this initial list can be stated as long as we want to keep the alternation between P and Q during the progress of our dialogue. If a is atomic, a sequent such as $a \Rightarrow a \vee w$ produces the initial list $Qa, Pa \vee w$ which *must* be followed by an attack of Q ; on the other hand, a sequent such as $a \wedge w \Rightarrow a$ produces the initial list $Qa \wedge w, Pa$ which *must* be followed by an attack of P . Certainly, regulations circumventing these difficulties may be formulated, but apparently only at the cost of a loss in intuitive appeal.

Obernau/Neckar

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EDITOR'S NOTE

The dialogue system of this chapter has recently been implemented as a theorem prover **DIALOG** [1] and **COLOSSEUM** [2]. **DIALOG**, written in Lisp, offers a rule language for redefining the rules of the game. It also supports automatic and interactive proving and has a user-friendly interface. Efficiency, however, has not been the major concern, and therefore, **DIALOG** may serve more as a tool for teaching or for experimenting with the dialogue rules.

COLOSSEUM is a no-frills re-implementation of Dialogue Games in Prolog. Its dialogue rules are hardwired for intuitionistic first order predicate logic (as specified by Felscher above). **COLOSSEUM** allows automatic proving only, but it is much faster than **DIALOG** and is web accessible.¹

¹<http://www8.informatik.uni-erlangen.de/IMMD8/staff/Zinn/Dialogue/Colosseum.html>

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